

Static yaw misalignment from high-resolution SCADA: the complete calculation, step by step

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Abstract

This note is the mathematical companion of the pipeline in `src/yaw/`. It describes every operation that is performed on the data, in the order in which the scripts perform it, with the exact thresholds that the code uses, and it derives the estimators from a single sensor model and a single aerodynamic law. Section 1 fixes the notation once; no symbol is introduced later without being defined where it first appears. Sections 2 to 6 follow the four pipeline steps (reconstruction and cleaning, consensus and states, physics estimators, submission), §7 states the scoring rule, and §8 collects the assumptions of every estimator in one table. Measured outcomes are quoted where they decide between competing models; the step reports `docs/01-docs/04` contain the full tables and figures.

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1 Notation

1.1 Conventions

All directions are compass bearings in degrees, clockwise from north. A difference of two bearings is always wrapped to $[-180^\circ, 180^\circ)$,

$$\text{wrap}(x) = ((x + 180) \bmod 360) - 180. \quad (1)$$

Angles enter trigonometric functions in radians; the conversion constant is written $\rho = \pi/180$ wherever it matters.

Time has three resolutions. The raw stream is sampled at roughly 12s; a raw sample is indexed by t . Minute aggregates are indexed by m (1-minute bins) or m_{10} (10-minute bins). Calendar days are indexed by d . Turbines are indexed by i (the target) and j (a neighbour). A contiguous run of days with constant behaviour is a *state*, indexed by ℓ .

For a set of bearings $x_1, \dots, x_n$ the *circular mean* and the *mean resultant length* are

$$\bar{x} = \text{atan2}\left(\frac{1}{n} \sum_i \sin x_i, \frac{1}{n} \sum_i \cos x_i\right), \quad \bar{R} = \left| \frac{1}{n} \sum_i e^{ix_i} \right| \in [0, 1], \quad (2)$$

where $\bar{R} = 1$ means all bearings are equal and $\bar{R} \rightarrow 0$ means they are spread around the circle. The *circular median* is the ordinary median of the wrapped deviations from the circular mean,

$$\text{med}^\circ(x) = \bar{x} + \text{med}_i \text{wrap}(x_i - \bar{x}). \quad (3)$$

It inherits the robustness of the median and does not depend on where the $\pm 180^\circ$ cut sits. med without subscript is the ordinary median.

1.2 Symbols

Table 1 lists every symbol used in the note. Physical quantities that carry the undisclosed scale factor of the data set (power, wind speed, rotor and generator speed, pitch) are only ever used in ratios, differences or threshold comparisons, never as absolute physical values.

2 Step 1: reconstruction, cleaning and aggregation

Code: `yaw.io`, `yaw.aggregate`, `yaw.quality`, `yaw.labels`, `yaw.geometry`; *script scripts/build_cache.py*; *report docs/01*.

2.1 The change-based stream

The parquet files hold one row per turbine and timestamp with seven channels (P , U , Ω , Ω_g , φ , N_r , W_r). The stream is *change-based*: a channel is written only when its value changed, and a null means “unchanged since the previous row”, not “missing”. Per channel the fraction of written rows is 0.5–0.8, except N_r , which is written on only $\approx 6\%$ of rows because the nacelle moves only during yaw manoeuvres.

Reconstruction rule. Rows are sorted by (turbine, t). Before filling, the indicator

$$u_t = \mathbf{1}[N_r \text{ written at } t] \quad (4)$$

is kept: it marks a *yaw controller action* and is lost after filling. Then every channel is forward-filled per turbine (`ffill`), so that each row carries the last known value of every channel. Nothing is interpolated; a value holds until the next write.

The vane. The vane angle is not shipped. In this SCADA the wind direction field is computed by the controller as heading plus vane, so

$$v_t = \text{wrap}(W_r - N_r)_t \quad (5)$$

returns the vane reading exactly (checked on the raw pairs in `docs/01`). Consequently W_r carries whatever offset the nacelle encoder carries, and only v is encoder-free. This single fact shapes the whole pipeline (§3).

Asynchronous writes (a trap). Because N_r and W_r are written independently, at a row where N_r was just updated the filled W_r still holds the value computed at the *previous* W_r write. Equation (5) then reads $v^{\text{old}} - \Delta$ (with Δ the heading change) until W_r refreshes, whatever the physical vane does. Nothing at the 12-second scale may therefore use v across a heading update; see §5.6.

2.2 Aggregation with circular statistics

The filled stream is binned to 1-minute and 10-minute bins. For a bin containing raw samples $t \in m$:

- linear channels ($P, U, \Omega, \Omega_g, \varphi, v$): the mean $\bar{P}_m$, etc.; in addition the within-bin standard deviation of P , U and v , and the median v_m^{med} of the vane;
- directions (N_r, W_r): the circular mean (2) and the resultant length $\bar{R}_m$; a naive mean of 350° and 10° would be 180° , and several turbines spend much time near north;
- counts: number of raw rows n_m , number of controller actions $\sum_{t \in m} u_t$, and the operating fraction $\omega_m = \frac{1}{n_m} \sum_{t \in m} \mathbf{1}[P_t > P_{\min}]$.

Per turbine this gives $\approx 1.0 \times 10^6$ minutes from $4.7\text{--}5.9 \times 10^6$ raw rows over the 727–731 days of record.

2.3 Operating regimes

The undisclosed scale factor forbids physical thresholds; the thresholds are therefore set from the data. The 99.9th percentile of P defines $P_{\text{rated}} = 4800$. A minute is

$$\text{operating} \iff \bar{P}_m > P_{\min} = 100 \quad (\approx 2\% \text{ of rated}), \quad (6)$$

$$\text{partial load} \iff \text{operating} \wedge \bar{P}_m < 0.85 P_{\text{rated}} \wedge \bar{\varphi}_m < 0.5^\circ. \quad (7)$$

Pitch is $-3 \dots -1.7^\circ$ below rated and rises above it, so the pitch condition removes the above-rated region cleanly. Above rated the controller holds power constant by pitching and the power carries no yaw information; every power-based estimator therefore works in partial load only. The estimators of §5 use a tighter version, (26).

2.4 Daily table and sensor-health flags

Per turbine and day the minutes are summarised: number of minutes, rows and controller actions, operating and partial-load minutes, and, over the *operating* minutes only, the daily median vane

$$v_d^{\text{med}} = \text{med}_{m \in d, \text{operating}} v_m^{\text{med}}, \quad (8)$$

its inter-quartile range, the mean within-minute vane standard deviation, the median pitch and rotor speed, and the circular mean heading. Three flags mark days that no estimator may use:

frozen vane : mean within-minute vane std $< 0.05^\circ \wedge$ operating minutes > 60 ,

no yaw : $\sum_d u_t = 0 \wedge$ operating minutes > 60 ,

sparse : raw rows < 1000 .

A day is *healthy* when none of the three is set. The flags catch two real faults: PPP_WTG13 reports $v = -34.0^\circ$ with zero variance and no controller action on 8–16 Feb 2024 and $v = +39.1^\circ$ on 9–12 Oct 2024 (11 and 7 days); no other turbine has a frozen vane. Sparse days are 1–8 per turbine.

2.5 Labels and label-derived states

The label θ_d^* is given per row of the training turbines and is constant within a day; the daily series is its first value per day (195, 312 and 195 labelled days on PPP_WTG12/13/14, January to November 2023). The organisers score only days outside a transition window around each label step, but do not publish the window rule; it is reconstructed as follows. The daily label series of each turbine is segmented into piecewise-constant states with the PELT algorithm (§4.4) under an L_2 cost with penalty $\lambda = 20$ and minimum length $n_{\min} = 7$ days. The state level is the median label within the state. A day is a *transition* when it lies within 7 calendar days of a state boundary (gaps count as days) or when its label deviates by more than 0.75° from its state level; a day is *scored* when it is labelled and not a transition. This gives 2, 6 and 2 states and 180, 217 and 180 scored days. All local scoring (§7) uses this table; the rule lives in one function (`yaw.labels.add_states`) so it can be changed in one place.

2.6 Layout, neighbours and wake sectors

The layout files give coordinates with an unknown common scale, so distances are only used as ratios and bearings are exact. For turbines i, j on the same site with coordinate differences $(\Delta x, \Delta y)$,

$$\phi_{ij} = \text{atan2}(\Delta x, \Delta y) \bmod 360 \quad (9)$$

is the bearing from i to j . The neighbours of i are the same-site turbines sorted by distance. A minute of turbine i is *waked* when its (de-stepped, §4.2) wind direction lies within $\pm 20^\circ$ of ϕ_{ij} for any published neighbour j closer than four times the distance of the nearest published neighbour. Only 16 of the site’s turbines are published, so the mask is a lower bound on real waking; residual partial wakes are one of the sector-dependent nuisances of Lemma 1.

2.7 Pair offsets

For every same-site pair (i, j) and every 10-minute bin in which both turbines operate more than 90 % of the bin ($\omega_{m_{10}} > 0.9$ for both), the heading difference $D_{ij} = \text{wrap}(N_r^i - N_r^j)$ and the wind-direction difference $\text{wrap}(W_r^i - W_r^j)$ are formed, and their daily medians and inter-quartile ranges are stored (65 pairs). This table is diagnostic ([figures/01_vane_and_offsets.png](#)); the consensus of §4 recomputes the pair series with a circular median.

2.8 What the cleaning revealed

Three facts from this step drive the design:

1. **The vane set-point is not zero.** On every PPP turbine the daily median vane in operation is $-2.7 \dots -3.7^\circ$; on SSS it is -0.8 or -3.1° , and SSS_WTG06 wanders between 0 and -4° over weeks. The controller nulls the vane at $\approx -3^\circ$, not at 0.
2. **Encoders lose north.** PPP_WTG12’s heading jumps by $\approx 138^\circ$ against every neighbour from August 2023 while its label does not move; on 10–17 % of days a turbine of the WTG12/13/14 cluster sits $100\text{--}180^\circ$ away from all neighbours for several days while producing normally. These are encoder-frame events, not flow.
3. **Labels and vane can move in opposite directions.** On PPP_WTG13 the vane median moved from ≈ -0.5 to $\approx -3^\circ$ on 14–18 January 2023 while the label went from -11.5 to -8.1° . This becomes the sign check of §6.5.

3 The sensor model

3.1 True and reported quantities

At any instant the unobserved quantities are the true wind direction W_t , the true nacelle heading N_t and the yaw error $\gamma = \text{wrap}(W_t - N_t)$. The stream reports

$$N_r = N_t + \delta_n, \tag{10}$$

$$v = g \gamma + \delta_v, \tag{11}$$

$$W_r = N_r + v. \tag{12}$$

Equation (10) says the encoder has no north reference: δ_n is an arbitrary constant that changes at every re-reference. Equation (11) says the vane measures the yaw error with a mounting offset δ_v and a gain $g \leq 1$; the gain models the well-documented damping of the inflow angle behind a rotating rotor (wake swirl, nacelle body).¹ Equation (12) is not an assumption but the encoding of the data set, §2.

¹Everything below is written for general g ; the implementation uses $g = 1$ because §5.6 shows g cannot be measured from this data.

3.2 The controller and the definition of the label

The yaw controller filters v over minutes and rotates the nacelle whenever the filtered reading leaves a deadband (measured occupancy $\approx \pm 8^\circ$) around its set-point s . In steady state the controller therefore enforces $\mathbb{E}[v] = s$, and from (11)

$$\theta \equiv \mathbb{E}[\gamma] = \frac{s - \delta_v}{g}. \quad (13)$$

θ is the *static yaw misalignment*: the persistent yaw error the controller cannot see because it sits inside its own measurement. The WindFit reference system measures it with an independent sensor and fits the power-optimal heading over several days, which is why its daily series is smoothed and drifts for about a week before each step, and why days near a step are not scored. The label is negative on all three training turbines; the sign check of §6.5 establishes that the label sign coincides with γ as defined here.

Two consequences shape everything that follows.

1. **The vane cannot see θ .** Averaged over any window the vane reads s whatever θ is. Measured: $v_d^{\text{med}} \approx -3^\circ$ on every PPP turbine irrespective of the label (correlation with the label -0.33). Information about θ must come from the *response* of the turbine to γ (power) or from an external direction reference (neighbours).
2. **θ changes in three ways** that the sensors do not treat alike: a change of the set-point s , of the vane offset δ_v , or of the gain g . All three move θ by (13); only the first two move the neighbour residual of §4; and an encoder re-reference (δ_n) moves the residual without moving θ at all.

3.3 Aerodynamic model of power under yaw

Actuator-disc theory for a rotor yawed by γ replaces the free-stream speed U_∞ by its axial component $U_\infty \cos \gamma$ in the mass flow and in the extracted energy, which gives $P(\gamma) = P(0) \cos^3 \gamma$. Real rotors show a smaller exponent because the disc also captures part of the tangential component and because the power coefficient shifts with γ under torque control. The standard empirical form, also assumed by the OpenOA static-yaw method, is

$$P(U_\infty, \gamma) = P_0(U_\infty) \cos^k \gamma, \quad k \in [1.4, 3], \quad (14)$$

with P_0 the power of the aligned rotor. The implementation fixes $k = 2$ after finding that a free k wanders between 0.2 and 13 from month to month (§5.2.1); with k fixed the only unknown of the curve is its location.

Small angles and the deadband. With γ in radians,

$$\cos^k \gamma = 1 - \frac{k}{2} \gamma^2 + O(\gamma^4), \quad \ln \cos^k \gamma = -\frac{k}{2} \gamma^2 - \frac{k}{12} \gamma^4 + O(\gamma^6). \quad (15)$$

Over the $\pm 8^\circ$ deadband, with $k = 2$, the power varies by $\frac{k}{2}(8\rho)^2 \approx 2\%$. Every estimator that locates the peak of (14) from operational data must therefore resolve a curvature whose full effect is two per cent of power, against sector-, turbulence- and season-dependent power variations of the same order. This is the root cause of the failures of C1 and C2.

The nacelle anemometer. The anemometer sits behind the rotor and reads $U = U_\infty h(\gamma)$ with $h(0) < 1$ and $\partial_\gamma h \neq 0$: the rotor-induced deficit itself depends on the yaw angle. Binning by U is therefore not binning by U_∞ ; the apparent power-yaw curve at fixed U is $\cos^k \gamma / h^3(\gamma)$. Since h is even in γ to leading order this does not move the peak, but it flattens or sharpens the curve, i.e. it is absorbed into an effective k : another reason to fix k .

4 Step 2: encoder de-stepping, consensus residual and states

Code: `yaw.consensus`; script `scripts/segment_states.py`; report `docs/02`.

4.1 Sensor algebra of the residual

Suppose a consensus estimate $c(t)$ of the true wind direction is available from the neighbours, of the form $c = W_t + \kappa$ with κ a constant that depends on the neighbours' own offsets. From (10) and $\gamma = W_t - N_t$,

$$\text{wrap}(N_r - c) = N_t + \delta_n - W_t - \kappa = -\gamma + \delta_n - \kappa. \quad (16)$$

Averaged over a day, $\mathbb{E}[\gamma] = \theta$, so the daily residual obeys

$$\boxed{r(d) = -\theta(d) + \delta_n(d) - \kappa} \implies \Delta r = -\Delta\theta \quad \text{whenever } \Delta\delta_n = 0. \quad (17)$$

Proposition 1. *The residual identifies changes of the static misalignment in physical degrees with slope exactly -1 , independent of the vane (g, δ_v) and of the aerodynamics; its level is not identifiable, because δ_n and κ are unknown constants.*

This is the ‘‘consensus anchor’’ of the leading team. Using W_r instead of N_r gives $W_r - c = \delta_n + \delta_v + (g - 1)\gamma - \kappa$, a constant for $g = 1$ and a damped version of (17) for $g < 1$; the implementation uses N_r . The residual is blind to changes of θ that occur through g (the wind side of γ does not move), but such changes are also invisible to the label mechanism to first order.

The consensus c is never formed explicitly. Instead the residual is built pair by pair, §4.3, which is equivalent because every pair difference $D_{ij} = \text{wrap}(N_r^i - N_r^j)$ contains $-\gamma^i + \gamma^j + \delta_n^i - \delta_n^j$ and the neighbour's terms are constant over a day up to noise.

4.2 Pair series and encoder re-references

δ_n is piecewise constant with jumps at re-references (service, power cycle of the encoder). A jump of δ_n^i moves D_{ij} against *all* neighbours j by the same amount on the same day; the detection rule uses exactly this signature.

Reference set. For each turbine i the $n_{\text{nb}} = 4$ nearest same-site turbines are its references, excluding PPP_WTG12 and PPP_WTG16, whose headings are too unstable to serve as references (they still receive their own residual from the remaining neighbours).

Daily pair series. From the 10-minute aggregates, for bins where both turbines operate more than 90 % of the time,

$$D_{ij}(d) = \text{med}_{m_{10} \in d}^\circ \text{wrap}(N_r^i - N_r^j)_{m_{10}}. \quad (18)$$

Level shifts. Each daily pair series is unwrapped (so that a crossing of $\pm 180^\circ$ is not read as a jump) and segmented with PELT under an L_1 cost, penalty $\lambda = 150$, minimum length $n_{\text{min}} = 10$ days (§4.4). The L_1 cost (median per segment) is chosen so that single outlying days do not create segments. Each boundary yields a candidate jump J , the wrapped difference of the medians of the adjacent segments.

Attribution rule. A candidate jump on date d in the series (i, j) is credited to the encoder of i when candidates with $|J| \geq 25^\circ$ of the same sign appear within ± 3 days in at least 60% of i 's pairs and in at least two of them. The credited jump is the median of the matching candidates and its date the earliest of them. Shifts below 25° cannot be attributed by majority: a 5° misalignment change and a 5° re-reference look identical in every pair. That is why the state segmentation later types boundaries with a jump above 15° as encoder events (§4.5) and the submission refuses to carry Δr across them.

De-stepping. With credited jumps $(d_1, J_1), (d_2, J_2), \dots$ of turbine i , both direction channels are corrected from each date on,

$$N_r^i(t) \leftarrow \left(N_r^i(t) - \sum_{n: d_n \leq t} J_n \right) \bmod 360, \quad \text{and likewise } W_r^i. \quad (19)$$

W_r must be corrected too because it carries δ_n by (12); v is unaffected. Sixty-seven steps on ten turbines were credited, none on PPP_WTG17.

4.3 Building the residual

After de-stepping, three cleaning operations and one correction are applied to each pair before the residual is formed.

1. **Excursion masking.** The daily pair series (18) is recomputed on the de-stepped headings and compared with its centred 31-day rolling median (at least 7 days present). Days with $|\text{wrap}(D_{ij}(d) - \text{rolling median})| > 30^\circ$ are dropped from the pair. These are the multi-day frame excursions shorter than the $n_{\min} = 10$ days of the step detector.
2. **Sector correction.** Terrain steering and partial wakes make the direction difference between two sites a function of the wind direction. With σ_j the 30° sector of the neighbour's (de-stepped) wind direction, $\sigma_j = \lfloor W_r^j / 30^\circ \rfloor$, the sector median $m_{ij}(\sigma_j) = \text{med}^\circ \{ D_{ij}(m_{10}) : \text{bins in sector } \sigma_j \}$ is subtracted bin by bin,

$$e_{ij}(m_{10}) = \text{wrap} (D_{ij}(m_{10}) - m_{ij}(\sigma_j(m_{10}))). \quad (20)$$

The neighbour's wind direction is used as the sector variable because it is not affected by the target's own yaw error.

3. **Daily pair residual.** $e_{ij}(d) = \text{med}_{m_{10} \in d} e_{ij}(m_{10})$ over days with at least 12 valid bins; its daily inter-quartile range is kept as a noise measure.
4. **Noisy references.** A pair whose median daily inter-quartile range exceeds twice the median of the turbine's pairs is dropped.

The daily residual of turbine i is the median over the surviving pairs,

$$r_i(d) = \text{med}_j e_{ij}(d). \quad (21)$$

The double median (over bins, then over neighbours) makes (21) robust to one bad neighbour and to a few bad bins; the measured day-to-day noise is $\approx 2^\circ$ (median absolute deviation). The price is that changes below $\approx 1^\circ$, the two small label steps on WTG12 and WTG14, are under the noise floor. In terms of §4.1, the sector medians m_{ij} absorb δ_n^j , θ^j and the terrain term into the constant κ , and r_i is (17) up to that constant.

4.4 Change-point detection: PELT

Both the label states (§2.5), the encoder steps (§4.2) and the SCADA states (§4.5) use the same algorithm. Given a series $y_1, \dots, y_T$, a partition into contiguous segments $\mathcal{S}_1, \dots, \mathcal{S}_M$ is chosen to minimise

$$\sum_{\ell=1}^M C(\mathcal{S}_\ell) + \lambda(M-1), \quad |\mathcal{S}_\ell| \geq n_{\min}, \quad (22)$$

with the segment cost either

$$C_{L_2}(\mathcal{S}) = \sum_{d \in \mathcal{S}} (y_d - \bar{y}_{\mathcal{S}})^2 \quad \text{or} \quad C_{L_1}(\mathcal{S}) = \sum_{d \in \mathcal{S}} |y_d - \text{med}(y_{\mathcal{S}})|. \quad (23)$$

The pruned exact linear time algorithm (PELT) solves (22) exactly by dynamic programming over the last change point, discarding candidates that can never become optimal. The L_2 cost is the negative log-likelihood of a piecewise-constant Gaussian mean up to scale, so λ acts as a BIC-type penalty: a break is accepted when it lowers the residual sum of squares by more than λ . For a level jump J over n days on both sides, the reduction is $nJ^2/2$, so with $\lambda = 400$ a jump is detected when $nJ^2/2 \gtrsim 400$, e.g. $J \geq 2.4^\circ$ over 140 days or $J \geq 5^\circ$ over 30 days. The penalties were chosen on the training turbines: lower values split PPP_WTG17's 2023 into up to 13 states and SSS_WTG06 into 50, noise rather than physics.

4.5 SCADA states

The residual is put on a complete daily calendar, smoothed with a centred 7-day rolling median (at least 3 days present) and linearly interpolated across gaps so that states are contiguous in calendar time,

$$\tilde{r}_i(d) = \text{med}\{r_i(d') : |d' - d| \leq 3\}. \quad (24)$$

PELT with the L_2 cost, $\lambda = 400$ and $n_{\min} = 14$ days is run on $\tilde{r}_i$. Each state ℓ receives the level

$$\bar{r}_\ell = \text{med}_{d \in \mathcal{S}_\ell} \tilde{r}_i(d). \quad (25)$$

The first day of each state is typed *encoder* when $|\bar{r}_\ell - \bar{r}_{\ell-1}| > 15^\circ$ (too large for a misalignment change) and *candidate* otherwise; days within ± 7 of a boundary are flagged as transitions, mirroring the scoring rule of §2.5.

Measured on the training turbines against the label-derived states of §2.5: adjusted Rand index 0.47 / 0.37 / 0.79 on WTG12/13/14, limited by the sub-degree label steps that r cannot see. After de-stepping, r tracks the inverted label at WTG13's two large events (20 January and 27 July 2023); at the July event $\Delta r \approx -9^\circ$ for a label step of $+3.65^\circ$, i.e. the service visit also re-referenced the encoder by a few degrees. *A small re-reference and a misalignment change are indistinguishable in r* : r gives boundaries reliably and magnitudes only as a prior. On PPP_WTG17 the result is two states with the boundary at 2024-01-06 and $\Delta r = -8.05^\circ$, i.e. $\Delta\theta = +8^\circ$ by (17); on SSS_WTG06, 13 states with three encoder-typed boundaries.

5 Step 3: physics estimators of the level

Code: yaw.estimators; script scripts/estimate_angles.py; report docs/03. The residual gives changes; the level must come from the turbine's response to γ . Four estimators were derived; none survived as a level estimator, and the derivations show why.

5.1 Common data selection

All estimators of this step use the 1-minute aggregates with the de-stepped wind direction. A minute of the target is *good* when

$$300 < \bar{P}_m < 3600, \quad \bar{\varphi}_m < 0.5^\circ, \quad \omega_m > 0.99, \quad \text{unwaked (§2.6),} \quad \text{healthy day (§2.4).} \quad (26)$$

The power window is narrower than (7): it excludes start-up and the knee below rated, where the $\cos^k$ law and the torque law both bend. When a reference neighbour is involved (C2, C4) the reference's minute must satisfy the first four conditions as well; the reference is chosen among the three nearest allowed neighbours as the one with the largest number of jointly good minutes.

Estimates are computed per *window*: each label state (scored days), each SCADA state (non-transition days), each calendar quarter, and the whole record.

5.2 C1: power-versus-vane peak with the nacelle anemometer

Estimator. Take good minutes with $|\bar{v}_m| < 20^\circ$ and a within-minute vane standard deviation below 6° , so that the minute has a well-defined yaw angle. The free-stream speed is unknown, so it is removed by normalisation within wind-speed bins: with $B(U)$ the 0.5 m/s bin of $\bar{U}_m$ (bins with fewer than 30 minutes are dropped),

$$\tilde{P}_m = \frac{\bar{P}_m}{\text{med}\{\bar{P}_{m'} : \bar{U}_{m'} \in B(\bar{U}_m)\}}. \quad (27)$$

By (14), and because the bin median is taken over all yaw angles in the bin, which are concentrated near the set-point,

$$\mathbb{E}[\tilde{P} \mid \bar{v}] \approx A \cos^k(\bar{v} - \theta_0), \quad \theta_0 = \delta_v \quad (g = 1), \quad (28)$$

where the peak location θ_0 is the vane reading at which the rotor is aligned, $\gamma = 0 \Leftrightarrow v = \delta_v$ by (11). For $g \neq 1$ the curve becomes $\cos^k((\bar{v} - \delta_v)/g)$: the peak is still at δ_v and the curvature is stretched by $1/g^2$, which with fixed k inflates A and leaves θ_0 unbiased to first order.

The minutes are binned by $\bar{v}$ in 1° steps; bin n is represented by its median y_n and its count c_n (bins with fewer than 10 minutes are dropped, and at least 8 bins are required). The two parameters are the weighted non-linear least-squares solution

$$(\hat{A}, \hat{\theta}_0) = \arg \min_{A, \theta_0} \sum_n c_n (A \cos^k(v_n - \theta_0) - y_n)^2, \quad |\theta_0| \leq 40^\circ, \quad (29)$$

found with a bounded trust-region method; in the free- k variant k is a third parameter with $0.2 \leq k \leq 15$. The bin median rather than the mean is used because the within-bin distribution of $\tilde{P}$ is heavy-tailed (start-up minutes, curtailment tails the power window does not fully remove).

From the peak to the misalignment. Combining (28) with the controller identity (13),

$$\hat{\theta} = \frac{s - \hat{\theta}_0}{g} \approx v^{\text{med}} - \hat{\theta}_0, \quad (30)$$

i.e. the misalignment is the distance between where the controller *holds* the vane and where the power *peaks* on the vane axis. The literature often reports $-\hat{\theta}_0$ alone, which silently assumes $s = 0$; on this fleet $s \approx -3^\circ$.

Sampling variance. Linearising the weighted residual vector $\mathbf{e}(p)$ of (29) at the solution with Jacobian $\mathbf{J}$ gives the Gauss–Newton covariance

$$\widehat{\text{Cov}}(\hat{p}) = \hat{\sigma}^2 (\mathbf{J}^\top \mathbf{J})^{-1}, \quad \hat{\sigma}^2 = \frac{\|\mathbf{e}(\hat{p})\|^2}{n_{\text{bins}} - \dim p}, \quad (31)$$

and the reported standard error is the square root of the θ_0 diagonal element. It is a conditional error: it assumes the bin medians scatter independently around a $\cos^k$ curve. The measured quarter-to-quarter scatter of $\hat{\theta}_0$ within one label state (4–9°) is five to ten times larger, which proves that model error dominates.

5.2.1 Why the peak is biased: the tilt lemma

Lemma 1 (Peak shift under a linear nuisance). *Let the true bin curve be $y(v) = A[1 - \frac{k}{2}\rho^2(v - \theta_0)^2]$ in degrees, and let an unmodelled nuisance add a linear trend εv (normalised power per degree). Then the peak of the observed curve is at*

$$v^* = \theta_0 + \frac{\varepsilon}{Ak\rho^2}. \quad (32)$$

Proof. $\partial_v[y(v) + \varepsilon v] = -Ak\rho^2(v - \theta_0) + \varepsilon = 0$. $\square$

With $k = 2$, $\rho^2 = 3.05 \times 10^{-4}$, $A \approx 1$, a tilt of one per cent of power across the 8° deadband, $\varepsilon = 0.01/8 = 1.25 \times 10^{-3}$ per degree, shifts the peak by $1.25 \times 10^{-3}/(2 \times 3.05 \times 10^{-4}) \approx 2^\circ$. Any effect that correlates power with the sign of the vane deviation at the one-per-cent level (a sector with slightly different shear or turbulence, or the fact that the vane deviates to one side while the wind is veering and veering periods have their own power statistics) moves the fitted misalignment by degrees. The measured curves change shape between quarters of the same label state (monotone rising, monotone falling, peaked; `figures/03_estimators.png`), which is this lemma at work. The same mechanism makes the free- k fit unstable: k and the tilt are nearly collinear on a $\pm 8^\circ$ support. This is the failure the challenge README warns about (“ranks the three training turbines in the wrong order”) and the one the leading team’s T0 entry reproduces.

5.3 C2: neighbour power as the wind reference

C2 keeps (28)–(29) but replaces the wind-speed bin in (27) by a bin of the *simultaneous power of the reference neighbour*, $\bar{P}_m^j$, in bins of 150 power units, using minutes that are good for both turbines:

$$\tilde{P}_m^{\text{C2}} = \frac{\bar{P}_m^i}{\text{med}\{\bar{P}_{m'}^i : \bar{P}_{m'}^j \in B(\bar{P}_m^j)\}}. \quad (33)$$

The motivation is the anemometer deficit $h(\gamma)$ of §3.3: the neighbour’s power is a wind reference that does not sit in the target’s wake and is not a function of the target’s yaw angle.

Assumption 1. (i) *The neighbour’s free-stream speed is a monotone function of the target’s, stable over the window;* (ii) *the neighbour’s own yaw error is stationary over the window.*

Under these the ratio has the same peak $\theta_0 = \delta_v$ and the derivation of C1 goes through. What C2 removes and what it adds is decided by Lemma 1: it removes the yaw dependence of the anemometer, which is even in γ and does not shift the peak anyway, and it adds the neighbour’s sector dependence (the speed map between two sites depends on wind direction, which correlates with the vane deviation). Empirically the C2 standard errors are 3–6° and its levels (−5, −18, −18° on the training turbines) are further from the labels than C1. The neighbour is the right idea for the *speed* bias and the wrong tool for the *tilt* bias.

5.4 C3: the rotor-speed method

C3 was planned and not implemented; the derivation shows why it adds nothing here. In partial load the generator torque follows the optimal tip-speed law

$$T_g = K_g \Omega^2, \quad P = T_g \Omega = K_g \Omega^3, \quad (34)$$

with K_g a constant of the controller. Rotor speed and power are therefore deterministically linked *independently of the yaw angle*: Ω against P is the same curve at every γ , and “rotor speed at a given power” carries no yaw information. Yaw information appears only when Ω is compared with an external speed reference: at the equilibrium of (34) with the aerodynamic torque, $\Omega \propto U_\infty \cos \gamma$ to leading order, so $\Omega/U = \text{const} \cdot \cos \gamma / h(\gamma)$ peaks at $\gamma = 0$. But this is the C1 estimator with $P^{1/3}$ in place of P and the same anemometer, and by (34) it is algebraically the same fit. The rotor-speed method is useful where the power signal is curtailed or corrupted and Ω is not; here the reverse holds.

5.5 C4: yaw manoeuvres as natural experiments

Idea. Every yaw manoeuvre rotates the nacelle by a *known* angle Δ , and a rotation is a rotation regardless of δ_n , δ_v or g . The power before and after is measured at two yaw errors that differ by a known amount, so the pair samples the *derivative* of the $\cos^k$ curve at a known offset from the operating point. The vane is not used; the estimate is in physical degrees. This is the only estimator of the pipeline whose level is, in principle, free of the vane.

Manoeuvres. From the filled 12-second stream, heading updates ($u_t = 1$) with $|\text{wrap}(\Delta N_r)| > 0.05^\circ$ are grouped into runs separated by less than 60 s. A run is a manoeuvre with start t_0 , end t_1 , number of updates, duration $t_1 - t_0$, and rotation

$$\Delta = \text{wrap}(N_r(t_1) - N_r(t_0^-)), \quad (35)$$

the heading after the last update minus the heading before the first. There are $\approx 10^5$ per turbine (ramps of $\approx 0.5^\circ/\text{s}$, $2\text{--}30^\circ$).

Events. A manoeuvre with $2^\circ \leq |\Delta| \leq 30^\circ$ and duration $\leq 300\text{ s}$ is an event if the five 1-minute bins from 6 to 2 minutes before t_0 and the five bins from 2 to 6 minutes after t_1 are all good for target and reference. Let $\ln P^\pm$ be the mean of $\ln \bar{P}_m$ over the after/before window of the target and $\ln P_j^\pm$ the same for the reference. The event statistic is the double log-ratio

$$\ln Q \equiv (\ln P^+ - \ln P^-) - (\ln P_j^+ - \ln P_j^-). \quad (36)$$

Between 9 and 26×10^3 events per turbine survive.

Derivation. Let the manoeuvre rotate the nacelle from N_t^- to $N_t^+ = N_t^- + \Delta$. The controller rotates towards the wind, so $\Delta > 0$ means the wind had drifted clockwise of the nacelle. After the manoeuvre the controller has restored its equilibrium and the yaw error is the static misalignment, $\gamma^+ = \theta$. Before it, the yaw error was θ plus the part of the drift that had already accumulated,

$$\gamma^- = \theta + f\Delta, \quad 0 < f \leq 1, \quad (37)$$

where $f = 1$ if the wind shifted as a step and the nacelle followed, and $f < 1$ if the wind drifted gradually and the “before” window already contains part of the drift. Under (14), with the free-stream speed change cancelling between target and reference (both in partial load, both unwaked),

$$\ln Q = k[\ln \cos \theta - \ln \cos(\theta + f\Delta)] + \eta. \quad (38)$$

Expanding $\ln \cos(\theta + x) = \ln \cos \theta - x \tan \theta - \frac{x^2}{2 \cos^2 \theta} + O(x^3)$ with $x = f\Delta$ in radians,

$$\ln Q = kf \tan \theta \Delta + \frac{kf^2}{2 \cos^2 \theta} \Delta^2 + O(\Delta^3) + \eta. \quad (39)$$

Proposition 2 (Sign and level from the manoeuvre slope). *The coefficient of the term linear in Δ is $\beta_1 = kf \tan \theta$ per radian. Its sign is the sign of θ in the convention $\gamma = W_t - N_t$, clockwise positive, and its magnitude gives*

$$\hat{\theta} = \arctan \frac{\hat{\beta}_1}{kf}. \quad (40)$$

The quadratic term is even in Δ and positive: a manoeuvre in either direction raises power on average, by more for larger $|\Delta|$. Only the asymmetry between $+\Delta$ and $-\Delta$ carries θ .

Intuitively, if $\theta < 0$ (rotor pointing clockwise of the wind), a manoeuvre with $\Delta < 0$ started from $\gamma^- = \theta + f\Delta$, further from zero, and the gain after it is large; a manoeuvre with $\Delta > 0$ started closer to zero or past it, and the gain is small or negative. For $\theta = -5^\circ$, $f = 1$, $k = 2$: $\Delta = -20^\circ$ gives $P^+/P^- = \cos^2(-5^\circ)/\cos^2(-25^\circ) = 1.21$ and $\Delta = +20^\circ$ gives $\cos^2(-5^\circ)/\cos^2(15^\circ) = 1.06$. This asymmetry (large gain after -20° , little after $+20^\circ$) is what PPP_WTG17 shows in 2023.

Estimation. Events with $|\Delta| \leq 20^\circ$ (at least 50 of them) enter a robust regression of $\ln Q$ on Δ with the Huber loss

$$\ell_M(e) = \begin{cases} \frac{1}{2}e^2 & |e| \leq M \\ M|e| - \frac{1}{2}M^2 & |e| > M \end{cases}, \quad M = 1.35 \text{ (in units of the residual scale)}, \quad (41)$$

because $\ln Q$ has heavy tails (gusts inside a four-minute window). The quadratic term is not fitted: with Δ symmetric around zero it is orthogonal to the linear term and is absorbed by the intercept and the residual. The slope is fitted per degree and converted to per radian in (40). Its standard error is the standard deviation of the slope over 200 bootstrap resamples of the events, and the standard error of $\hat{\theta}$ follows from

$$\widehat{\text{se}}(\hat{\theta}) = \frac{1}{2}|\hat{\theta}(\hat{\beta}_1 + \text{se}) - \hat{\theta}(\hat{\beta}_1 - \text{se})|. \quad (42)$$

Remark 1 (Sign in the implementation). The function `manoeuvre_fit` originally returned the value $\arctan(-\hat{\beta}_1/(kf))$, which has the opposite sign to (40). Under the conventions of §3 (both headings clockwise bearings, $\Delta =$ after minus before) the derivation gives the positive sign; the code, the cached estimates, figure 03 and docs/03 were corrected on 2026-09-11. With the right sign C4 gives $+1.5$, -0.4 and -3.1° on the training turbines (labels -2 , -8 , -2°), and its 2023 to 2024 change on PPP_WTG17 is $\approx +5^\circ$, the same sign as the residual's $\Delta\theta = -\Delta r \approx +8^\circ$. This does not rescue C4 as a level estimator.

5.5.1 Why C4 is not the clean estimator hoped for

Two systematic effects, both measured, break Proposition 2 on this data.

1. **A meteorological confounder.** A *null test* applies the manoeuvre times of PPP_WTG17 to the power ratio of two unrelated turbines that did not manoeuvre. The slope should be zero; it is $-1.0 \pm 0.2 \times 10^{-3}$ per degree, a third of the “real” slope. The sign of Δ is the sign of the wind-direction change (veering versus backing), and veering and backing episodes come with their own power evolution (frontal passages, diurnal cycle) that is not identical at two sites a few hundred metres apart. The reference normalisation in (36) removes the common part of the speed change, not the direction-correlated residual. This term adds to β_1 directly and cannot be separated from θ without a model per sector.

2. **The unknown fraction f .** The level (40) scales as $1/f$, and f is neither known nor constant. With gradual drift and a $\pm 8^\circ$ deadband f is plausibly 0.3–0.6, so the raw slope underestimates $|\theta|$ by a factor two to three.

Different references disagree by $2\text{--}4^\circ$ and the three training turbines are not ranked correctly. C4 is kept as a feature (the asymmetry is real and large on some turbine-years) but not used for levels. The one open route to a label-free level is to difference the slope against the null distribution per wind-direction sector and manoeuvre direction.

5.6 C5: the vane gain, and why it cannot be measured here

If the vane responded instantaneously with gain g , a nacelle rotation Δ during steady wind would produce a vane jump of $-g\Delta$ ((11) with $\gamma \rightarrow \gamma - \Delta$), and the thousands of manoeuvres would give g directly, converting all vane-unit estimates into physical degrees. On this data the derived vane jumps by *exactly* $-\Delta$ at every heading update. The reason is the asynchronous encoding described in §2: at a heading update W_r still holds $N_r^{\text{old}} + v^{\text{old}}$, so (5) reads $v^{\text{old}} - \Delta$ until W_r refreshes, whatever the vane does. The apparent gain is one by construction, and no quantity derived from the two direction channels at the 12-second scale can recover g . C5 was dropped and $g = 1$ is used throughout.

6 Step 4: the submission model

Code: `yaw.submission`, `yaw.evaluate`; *script* `scripts/build_submission.py`; *report* `docs/04`.

6.1 Structure

For every turbine-day the prediction is

$$\hat{\theta}(d) = L + a + b x(d), \quad x(d) = \bar{r}_{\ell(d)} - \alpha_{\text{seg}(\ell(d))}, \quad (43)$$

where $\ell(d)$ is the day’s SCADA state (§4.5), $\text{seg}(\ell)$ its *encoder-free segment* (a maximal run of states without an encoder-typed boundary, i.e. the cumulative count of encoder boundaries up to ℓ), α the segment anchor, L the level prior, and (a, b) either fixed by physics or fitted on the labelled turbines:

T0 (label-free): $a = 0, b = -1$ from (17),

T3 (calibrated): (a, b) from §6.4.

The submitted `cluster` field is $\ell(d)$. Days without a SCADA state (gaps at the record edges) take the nearest state’s values; predictions are rounded to three decimals. The clusters are computed from the residual over all 731 days without any knowledge of the template or of which days are scored; the only calendar prior is $n_{\min} = 14$ days.

6.2 The level prior L

No label-free level estimator survived §5. The prior is the *vane set-point*: from (13) with $g = 1$ and the assumption $\delta_v = 0, \theta = s$, and s is measured as

$$L = \text{med}_d \text{ healthy } v_d^{\text{med}}, \quad (44)$$

the median over healthy days of the daily median vane in operation (8). On the PPP site this is -2.7 to -3.7° on every turbine, on SSS -0.8 to -3.3° . The prior is right when the vane is mounted true and the operator has chosen a non-zero set-point deliberately; it is wrong by $-\delta_v$ otherwise. On the training turbines it gives biases of $-0.45, +2.5$ and -0.8° : within a degree on two of three, and 4.5° off on WTG13, whose vane evidently carries its own offset.

6.3 The anchor α

The prior (44) is a statement about the turbine’s *typical* state, so the state levels must be centred before the slope is applied. Two choices were evaluated on the training turbines, both taken over the days d of the segment:

$$\alpha^{\text{med}} = \text{med}_{d \in \text{seg}} \bar{r}_{\ell(d)} \quad (\text{the level of the state holding most days}), \quad (45)$$

$$\alpha^{\text{mean}} = \frac{1}{|\text{seg}|} \sum_{d \in \text{seg}} \bar{r}_{\ell(d)} \quad (\text{day-weighted average of the state levels}). \quad (46)$$

The median anchor scores better on train (RMSE 2.18 versus 2.48) because the training turbines have one dominant state. For PPP_WTG17, whose two states hold 370 and 361 days, the median anchor is a coin flip that moves the whole prediction by 4° ; the mean anchor minimises the expected squared error under ignorance of which state is typical, and it was submitted first. Re-anchoring per segment is what stops re-references below the 25° attribution threshold of §4.2 from propagating into the levels.

6.4 T3 calibration

On the scored training days the label minus the prior, $y_d = \theta_d^* - L$, is regressed on x_d with the Huber loss (41),

$$(\hat{a}, \hat{b}) = \arg \min_{a,b} \sum_d \ell_{1.35}(y_d - a - b x_d), \quad (47)$$

giving $\hat{a} = -0.73^\circ$, $\hat{b} = -0.93$ on all three turbines: the labels confirm the physical slope of -1 and shift the level by less than a degree. Leave-one-turbine-out the fit is *worse* than T0 (RMSE 4.4 versus 2.5), because only WTG13 has label steps large enough to identify b ; holding it out collapses $\hat{b}$ to zero. The T3 file is therefore the T0 file with a -0.7° shift and a 7% shrink of the deltas, and the honest reading is that three labelled turbines cannot calibrate more than the sign.

6.5 Sign checks

Two independent tests are asserted before any file is written; the script aborts if either fails.

1. The least-squares slope of the label on x over the scored training days must be negative; measured -0.74 . This is (17) seen from the labels.
2. On PPP_WTG13, comparing operating samples before 13 January and after 19 January 2023, the median vane moved by -2.6° while the label moved by $+3.0^\circ$. By (13) a change of the set-point (or of δ_v with the controller re-nulling) moves θ by the same amount in the *same* direction, unless the label sign is opposite to the raw vane convention. The observed opposite motion therefore fixes the label as $-v$ in the raw convention, equivalently as $\gamma = W_t - N_t$ once the direction in which this controller reports the vane is accounted for. The test is asserted as “vane and label move in opposite directions” and requires $|\Delta v| > 1^\circ$ and $\Delta \theta^* > 1^\circ$.

Together the two tests pin the sign of the submission model, which is all that matters for scoring.

7 Scoring rule and what it rewards

7.1 Metrics

With $\mathcal{D}$ the set of scored days (§2.5),

$$\text{RMSE} = \sqrt{\frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} (\hat{\theta}_d - \theta_d^*)^2}, \quad \text{MAE} = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} |\hat{\theta}_d - \theta_d^*|, \quad \text{bias} = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} (\hat{\theta}_d - \theta_d^*). \quad (48)$$

The tie-breaker is the adjusted Rand index between the submitted clusters and the label-derived states, averaged over turbines. For a contingency table n_{uw} (cluster u , true state w) with margins $n_{u\cdot}$, $n_{\cdot w}$ and $n = |\mathcal{D}|$, Write $I = \sum_{uw} \binom{n_{uw}}{2}$ for the number of day pairs that share both a cluster and a true state, $A = \sum_u \binom{n_{u\cdot}}{2}$ and $B = \sum_w \binom{n_{\cdot w}}{2}$ for the pairs sharing a cluster and sharing a state respectively, and $E = AB/\binom{n}{2}$ for the value of I expected from a random partition with the same margins. Then

$$\text{ARI} = \frac{I - E}{\frac{1}{2}(A + B) - E}, \quad (49)$$

which is 1 for identical partitions up to relabelling and 0 in expectation for a random partition. It is invariant to cluster names; splitting a true state costs ARI in proportion to the split.

7.2 Effective sample size and the block bootstrap

Scored days are runs of a piecewise-constant series, and the error of a piecewise-constant prediction is constant within each run, so the effective number of independent observations is the number of states (two on PPP_WTG17, of order fifteen on the training set), not the number of days. The organisers resample whole states with replacement, score every submission on the same draw, and call a submission better than another only if its RMSE is lower on essentially all draws. The local harness reproduces this with 1000 draws over (turbine, state) blocks; the resulting 95 % interval for the T0 model on the training turbines is 1.8–3.3° around 2.48, which is the honest uncertainty of every score quoted in this note.

7.3 Implication for the strategy

RMSE differences smaller than that band are ties, and ARI then decides. The level of θ is what none of the physics estimators can pin better than a few degrees, so the winnable quantity is the segmentation (ARI, from §4) plus a level that is not badly wrong. The label-free T0 model scores 2.5° on the training turbines against a best-constant baseline of 3.1–6.1°; the current T0 leader on the public board scores 3.9° with a year-static segmentation.

8 Summary of assumptions per estimator

	measures	units	needs	dominant systematic
C1	peak of $\tilde{P}$ vs v at fixed U	vane (δ_v, g)	k fixed	tilt (Lemma 1), anemometer
C2	peak of P^i/P^j vs v	vane	healthy neighbour	tilt via neighbour sector
C3	Ω vs P	—	—	degenerate under torque law
C4	asymmetry of $\ln Q$ in Δ	physical	k, f	veering confounder, unknown f
C5	vane jump at Δ	—	synchronous channels	impossible (encoding)
r	$\Delta\theta$ from Δr	physical, slope —1	stable δ_n	re-references < 25°
L	vane set-point s	physical if $\delta_v = 0$	—	δ_v (WTG13: 4.5°)

<i>Reported SCADA channels (per raw sample t)</i>	
P	active power (scaled units; rated $P_{\text{rated}} \approx 4800$)
U	nacelle-anemometer wind speed (scaled)
Ω, Ω_g	rotor and generator speed (scaled)
φ	blade pitch angle (degrees; sits at $-3 \dots -1.7^\circ$ in partial load)
N_r	reported nacelle heading, field <code>NacDir</code>
W_r	reported wind direction, field <code>WindDir</code>
v	reported vane angle, derived: $v = \text{wrap}(W_r - N_r)$
<i>Physical (unobserved) quantities</i>	
W_t	true free-stream wind direction at the rotor
N_t	true nacelle heading (rotor axis)
γ	instantaneous yaw error $\gamma = \text{wrap}(W_t - N_t)$, positive when the wind comes from clockwise of the rotor axis
θ	static yaw misalignment $\theta = \mathbb{E}[\gamma]$ over a day or a state; the label
δ_n, δ_v	mounting offsets of the nacelle encoder and of the wind vane
g	gain of the vane ($g \leq 1$); the implementation uses $g = 1$
s	vane set-point of the yaw controller
<i>Aerodynamics</i>	
k	exponent of the $\cos^k \gamma$ power law; fixed at $k = 2$
$h(\gamma)$	rotor-induced speed deficit at the nacelle anemometer, $U = U_\infty h(\gamma)$
U_∞	free-stream wind speed
<i>Consensus and states</i>	
D_{ij}	heading difference of a pair, $D_{ij} = \text{wrap}(N_r^i - N_r^j)$
J	size of a level shift in a pair series
$m_{ij}(\cdot)$	sector median of D_{ij} in the neighbour's wind-direction sector
κ	unknown constant of the consensus (§4.1)
$r_i(d)$	daily consensus heading residual of turbine i
$\tilde{r}_i(d)$	its 7-day rolling median
$\bar{r}_\ell$	level of state ℓ (median of $\tilde{r}$ over the state)
$\lambda, n_{\min}$	change-point penalty and minimum state length of PELT
<i>Estimators</i>	
$\tilde{P}$	power normalised within a wind-reference bin (C1, C2)
A, θ_0	amplitude and peak location of the fitted $\cos^k$ curve
ε	slope of an unmodelled linear nuisance (tilt lemma)
Δ	signed nacelle rotation of a yaw manoeuvre (after minus before)
f	fraction of the wind shift already present in the “before” window of a manoeuvre
Q	double power ratio of a manoeuvre, (36)
β_1	slope of $\ln Q$ on Δ
η	noise term
ℓ_M	Huber loss with threshold M , (41)
<i>Submission</i>	
L	level prior of a turbine (vane set-point)
α	anchor of an encoder-free segment
x	centred state level $x = \bar{r}_\ell - \alpha$
a, b	offset and slope of the submission model
ϕ_{ij}	bearing from turbine i to turbine j

Table 1: Symbols. Hats denote estimates, bars denote means or levels.